

Computers and computing

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About...

- Computing
- Computation
- Computers
- Computer science

What is a computing?

Computing

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- Computing can be carried out by a person or some tool (machine)

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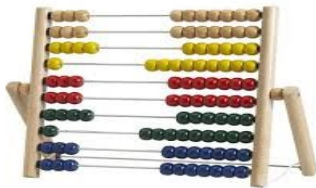
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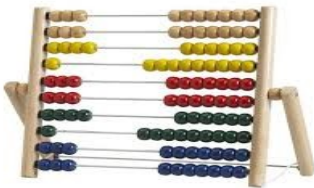
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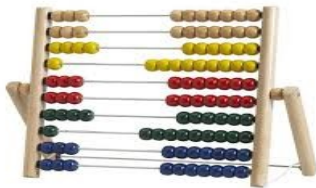
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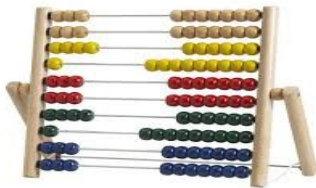
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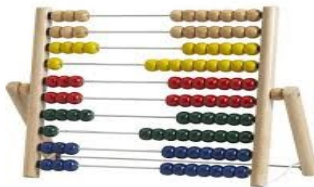
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Computing



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What is a computation?

Computation

- ♦ A computation (calculation) is a sequence of well-defined operations that uses some data as input (starting point) and produces a desired (and expected!) outcome
- ♦ The idea that problem can be solved by following a detailed set of instructions is very old
 - ♦ Ancient Greek, Persian and Chinese philosophers all developed computational approaches for solving arithmetic problems

What is a computer?

What is a computer?

Until the 1940s the word “computer” was a job title

Computers were people trained to follow a precise set of instructions required to create mathematical tables used for navigation, astronomy etc.

Computer

- ♦ (Modern) computer is an electronic machine that accepts data as an input, processes the input data (perform a computation), and produces output information

The computer is able to process input data to produce the desired output because there are instructions in its memory precisely describing the desired computation

- ♦ The usefulness of the modern computer is in the ability to process (and store) large amounts of data

The Brief History of Computers

- ♦ 2500 BC, Mesopotamia: Abacus
 - ♦ A table of successive columns which delimited the successive orders of magnitude
- ♦ 100 BC, Antikythera mechanism
 - ♦ Discovered in 1901. in the ship wrech near the Greek islan of Antikythera, it is belived to be the earliest known mechanical analog computer designed to calculate astronimic positions
- ♦ Mediavel Islamic world: number of mechanical devices that can be programmed to perform a specific task (Astrolab)

The Brief History of Computers

- ♦ 1642. Blaise Pascal designed a computer that added numbers entered with dials
 - ♦ The first signs of technophobia: mathematicians fearing the loss of their jobs!
- ♦ 1837. Charles Babbage describes the Analytical Engine which is accepted as the first design for a modern computer (it was never built)
 - ♦ Expandable memory, arithmetic unit, logic processing capabilities, able to interpret set of instructions with loops and conditional branching.
 - ♦ Turing complete!

The Brief History of Computers

- ♦ 1703. Binary logic developed by Gottfried Wilhem Leibnitz
 - ♦ Formal numerical system base on *ones* and *zeroes* which also represent *true* or *false* and *on* or *off* state
- ♦ 1854. George Boole published Boolean Algebra
 - ♦ Complete system that allowed computational processes to be mathematically modeled.
- ♦ 1936. Alan Turing and Alonzo Church introduced the formalization of an algorithm
 - ♦ Turing introduced Turing Machine, capable of simulating any computer algorithm!

The Brief History of Computers

- ♦ 1936. Z1 Computer built by Konrad Zuse, electrically driven mechanical calculator
- ♦ 1942. EINAC (Electrical Numerical Integrator and Calculator)
 - ♦ 18000 vacume tubes, capable of multiply two numbers at a rate of 300 per second
- ♦ 1946. John von Neumann architecture
 - ♦ Provided uniformity in subsequent computer designs

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Computers and Computing

- ♦ **Computing:** a process
- ♦ **Computation:** the sequence of steps to achieve the desired result
- ♦ **Computer:** the tool (slightly more advance than pencile and paper)

Computers and Computing

- Computing is much more fundamental than computers
- Computing may be done without a computer
- Computer cannot do much besides computing

What is a Computer science?

What is a Computer science?

- ♦ Science about computers?

What is a Computer science?

- Science about computers?

"Computer science is no more about computers than astronomy is about telescopes."

Edsger Wybe Dijkstra

Computer science

- ♦ Computer science is the scientific and practical approach to computation and its applications.
- ♦ It is systematic study of the feasibility, structure, expression and mechanization of the methodical processes (algorithms) that underlie the aquisition, representation, processing, storage, communication and access to information, whether such information is encoded in bits in a computer memory or in protein structures in a human cell.